INCH-POUND
MIL-B-49450/5A(ER)
8 November 1994

SUPERSEDING
MIL-B-49450/5(ER)
16 January 1987

# MILITARY SPECIFICATION SHEET

## BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT, BB-716/A, HARNESS ASSEMBLY

This specification is approved for use by the Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-B-49450(ER)

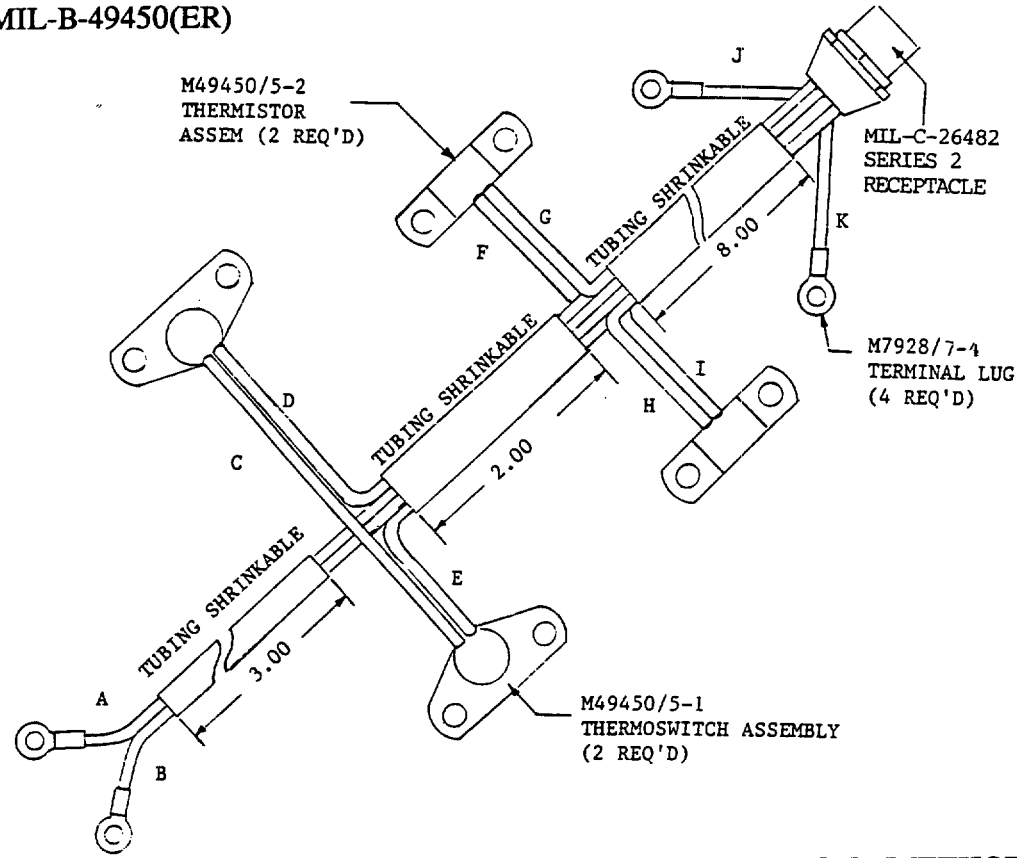



FIGURE 1. HARNESS CONFIGURATION (METHOD 1).


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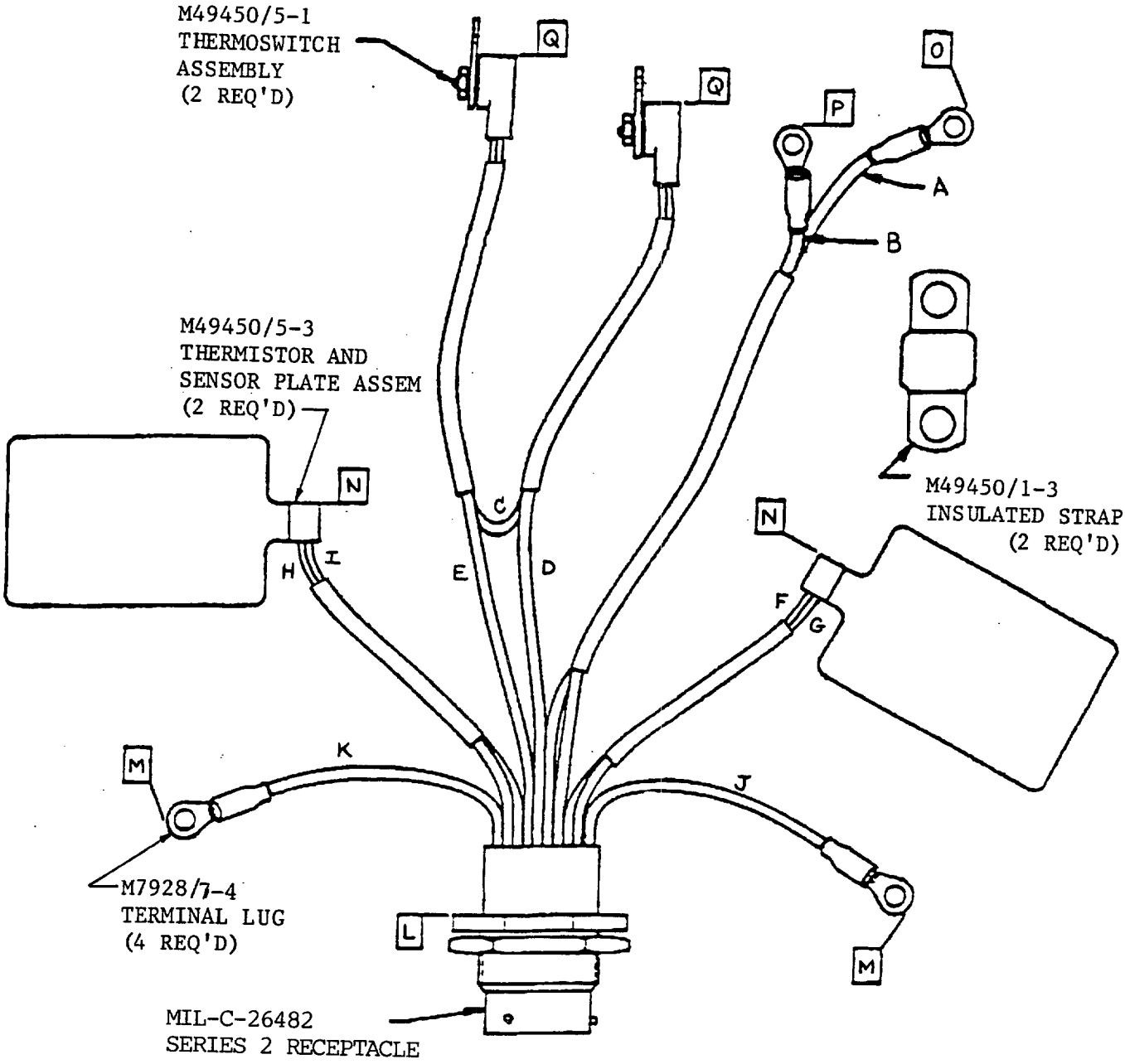



FIGURE 2. HARNESS CONFIGURATION (METHOD 2).



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TABLE 1. REFERENCE WIRE LISTING FOR FIGURE 1.

| REF LTR | WIRE DESCRIPTION |
|---|---|
| A | WIRE, BLUE TO PIN C, 18 INCHES LONG |
| B | WIRE, GREEN TO PIN H, 16.5 INCHES LONG |
| C | WIRE, WHITE, 2.5 INCHES LONG |
| D | WIRE, WHITE TO PIN E, 14 INCHES LONG |
| E | WIRE, WHITE TO PIN F, 12 INCHES LONG |
| F | WIRE, YELLOW TO PIN A, 10 INCHES LONG |
| G | WIRE, YELLOW TO PIN B, 10 INCHES LONG |
| H | WIRE, ORANGE TO PIN K, 10 INCHES LONG |
| I | WIRE, ORANGE TO PIN J, 10 INCHES LONG |
| J | WIRE, RED TO PIN G, 6 INCHES LONG |
| K | WIRE, BLACK TO PIN D, 6 INCHES LONG |

Table 2. Reference Wire Listing for Figure 2.

| REF LTR | WIRE DESCRIPTION |
|---|---|
| A | WIRE, BLUE TO PIN C, 16.0 ± .50 INCHES LONG FROM L TO O |
| B | WIRE, GREEN TO PIN H, 15.3 ± .50 INCHES LONG FROM L TO P |
| C | WIRE, WHITE AS REQUIRED |
| D | WIRE, WHITE TO PIN E, 14.38 ± .50 INCHES LONG FROM L TO Q |
| E | WIRE, WHITE TO PIN F, 14.38 ± .50 INCHES LING FROM L TO Q |
| F | WIRE, YELLOW TO PIN A, 9.12 ± .50 INCHES LONG FROM L TO N |
| G | WIRE, YELLOW TO PIN B, 9.12 ± .50 INCHES LONG FROM L TO N |
| H | WIRE, ORANGE TO PIN K, 9.12 ± .50 INCHES LONG FROM L TO N |
| I | WIRE, ORANGE TO PIN J, 9.12 ± .50 INCHES LONG FROM L TO N |
| J | WIRE, RED TO PIN G, 6.50 ± .50 INCHES LONG FROM L TO M |
| K | WIRE, BLACK TO PIN D, 6.50 ± .50 INCHES LONG FROM L TO M |



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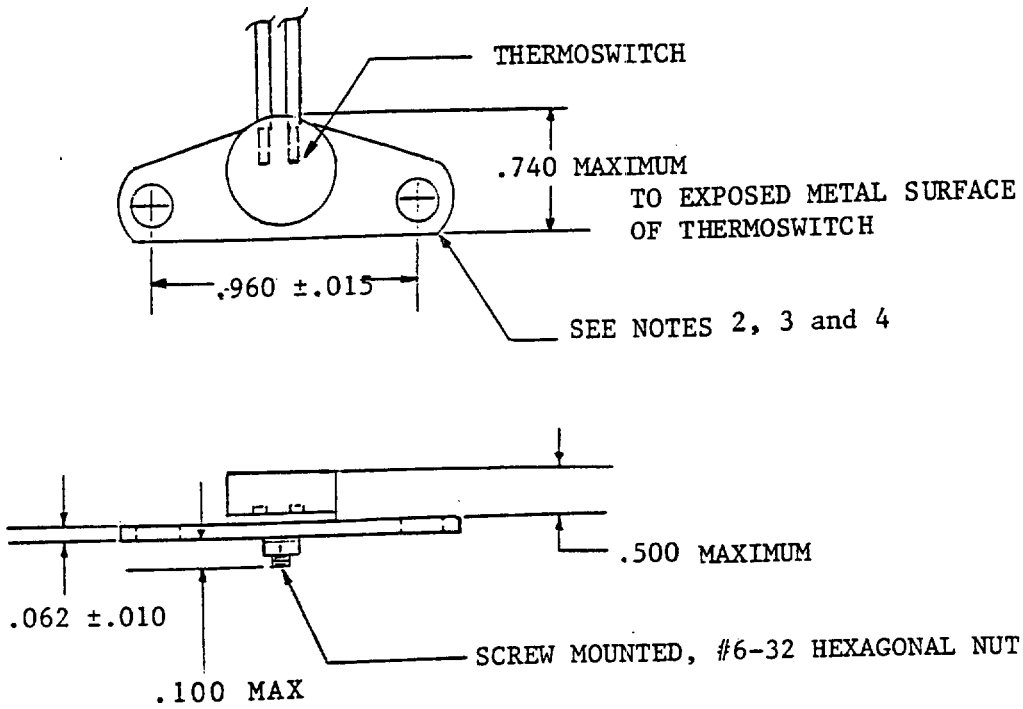



FIGURE 3. M49450/5-1 THERMOSWITCH ASSEMBLY.

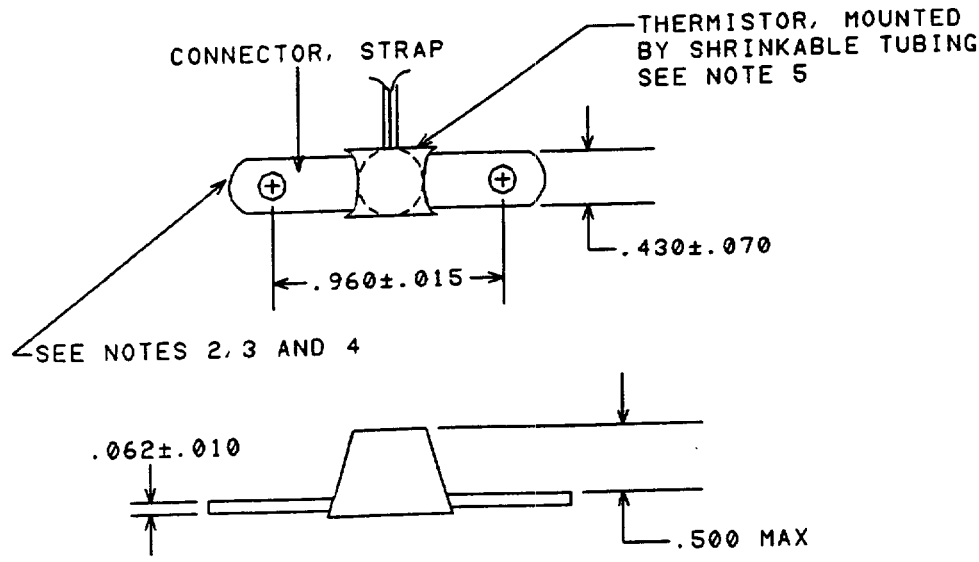



FIGURE 4. M49450/5-3 THERMISTOR ASSEMBLY (METHOD 1).



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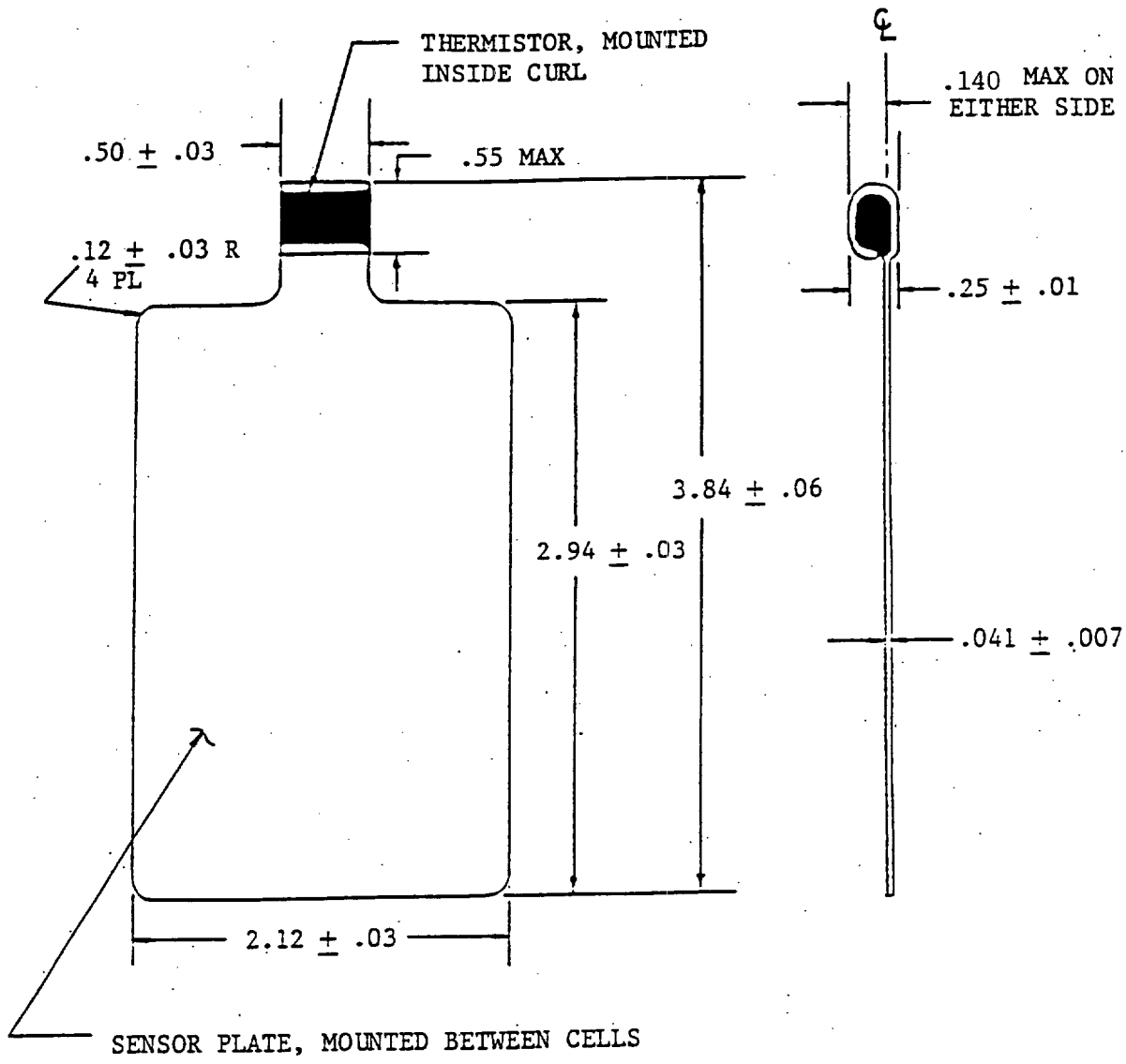



FIGURE 5. M49450/5-3 THERMISTOR ASSEMBLY WITH SENSOR PLATE (METHOD 2).



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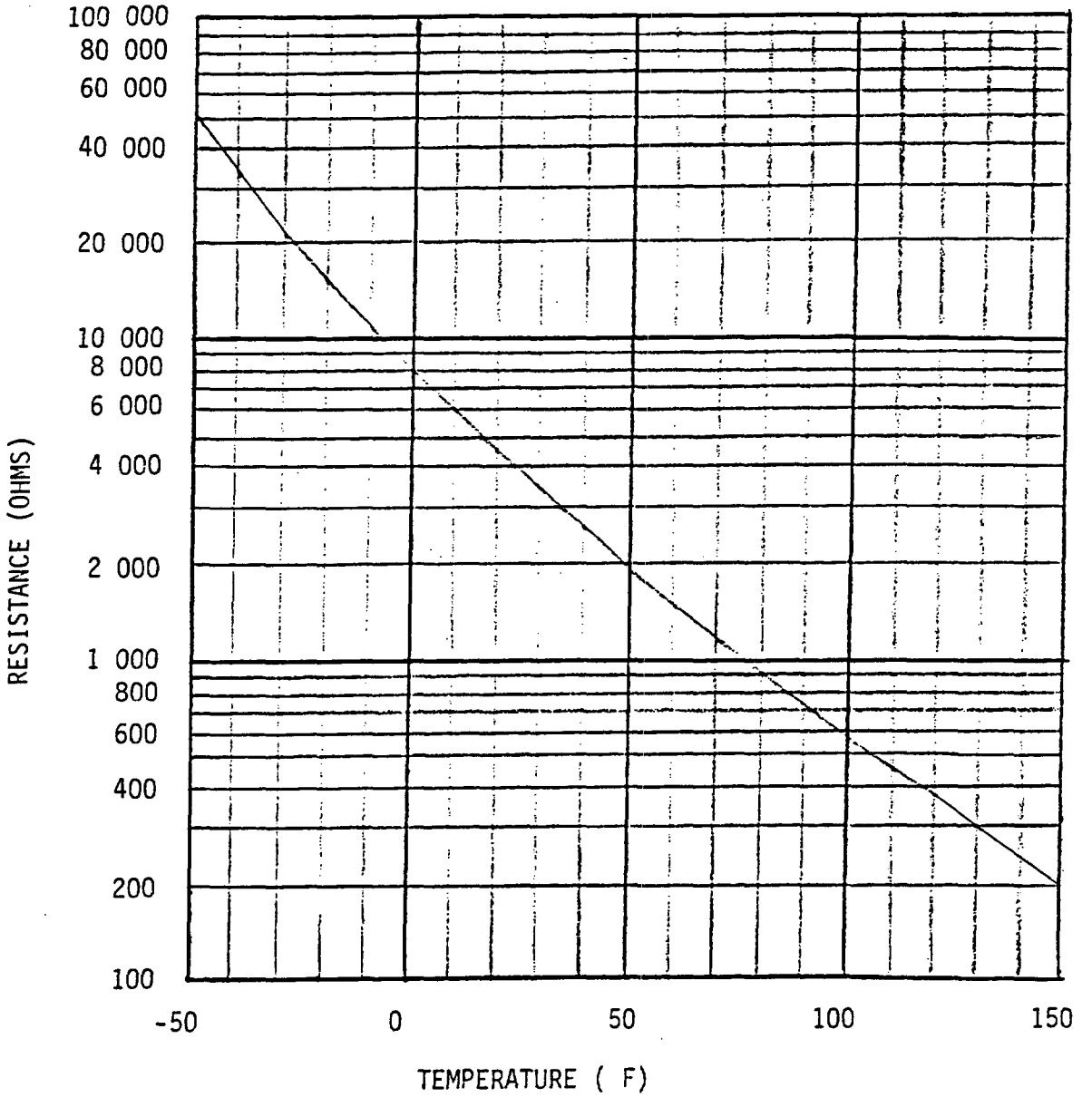



FIGURE 6. THERMISTOR OR TEMPERATURE DEVICE RESISTANCE-TEMPERATURE CHARACTERISTICS.



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DRAWING NOTES:

1. All dimensions are in inches, unless otherwise specified.

2. Strap shall be constructed of copper alloy conforming to ASTM-B133, ASTM-B272 or ASTM-B152.

3. Strap shall be plated in accordance with QQ-N-290, Class I, Grade E.

4. Strap shape is optional.

5. Thermistor may also be mounted to connector strap by screw mounting as per M49450/5-1 or equal method.

REQUIREMENTS:

1. The harness assembly shall use 22 AWG, teflon insulated wires conforming to MIL-W-16878 and resistant to the battery electrolyte. The length and color of wires shall be in accordance with Figure 1 and Table 1 or Figure 2 and Table 2.

2. All electrical connections made to the MS3474-L12-10PN receptacle and harness devices (thermistors, thermoswitches or equal) shall be potted with a compound that will not support flame, absorb moisture, or be damaged by battery electrolyte. Thermoswitch shall be modified to achieve maximum height of Figure 3.

3. The thermistor or temperature sensor device shall exhibit the resistance temperature characteristics of Figure 6.

4. The harness assembly shall be capable of withstanding the temperature and exposure of the battery electrolyte without damage.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing Activity:
Army-ER

(Project 6140-A800)



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